knowing nothing about us, might nevertheless conclude we're listening to.

Hydrogen is by far the most abundant kind of atom in the Universe. It's distributed in clouds and as diffuse gas throughout interstellar space. When it acquires energy, it releases some of it by giving off radio waves at a precise frequency of 1420.405751768 megahertz. (One hertz means the crest and trough of a wave arriving at your detection instrument each second. So 1420 megahertz means 1.420 *billion* waves entering your detector every second. Since the wavelength of light is just the speed of light divided by the frequency of the wave, 1420 megahertz corresponds to a wavelength of 21 centimeters.) Radio astronomers anywhere in the Galaxy will be studying the Universe at 1420 megahertz and can anticipate that other radio astronomers, no matter how different they may look, will do the same.

It's as if someone told you that there's only one station on your home radio set's frequency band, but that no one knows its frequency. Oh yes, one other thing: Your set's frequency dial, with its thin marker you adjust by turning a knob, happens to reach from the Earth to the Moon. To search systematically through this vast radio spectrum, patiently turning the knob, is going to be very time-consuming. Your problem is to set the dial correctly from the beginning, to choose the right frequency. If you can correctly guess what frequencies that extraterrestrials are broadcasting to us on—the "magic" frequencies—then you can save yourself much time and trouble. These are the sorts of reasons that we first listened, as Drake did, at frequencies near 1420 megahertz, the hydrogen "magic" frequency.

Horowitz and I have published detailed results from five years of full-time searching with Project META and two years of follow-up. We can't report that we found a signal from alien beings. But we did find something puzzling, something that for me in quiet moments, every now and then, raises goose bumps:

Of course, there's a background level of radio noise from Earth—radio and television stations, aircraft, portable telephones, nearby and more distant spacecraft. Also, as with all radio receivers, the